

Efficient Second Harmonic Generation in a Reverse-Polarization Dual-Layer Crystalline Thin Film Nanophotonic Waveguide

Lei Wang,* Xiuquan Zhang, and Feng Chen*

Second harmonic generation (SHG), one of the most significant $\chi^{(2)}$ nonlinear optical processes, plays a crucial role in a wide variety of optical and photonic applications. Designing various delicate schemes to achieve highly efficient SHG has become a long-standing and challenging topic in nonlinear optics. Despite numerous success on SHG based on birefringent phase-matching and quasi-phase-matching, so far, modal phase-matching (MPM) for SHG in tightly light-confined structures has still in its infancy. Here, a new scheme is proposed to realize efficient SHG via MPM by using a nanophotonic LiNbO_3 thin-film waveguide consists of two bonded layers with internally reversed polarizations. In such a dual-layer ridge waveguide based on lithium niobate on insulator (LNOI), upon optical excitation at 1574.6 nm, SHG is observed at 787.3 nm with a high conversion efficiency of $5540\% \text{ W}^{-1} \text{ cm}^{-2}$ experimentally. This work advances the understanding on modal phase-matched SHG and other quadratic optical nonlinear processes, offering additional strategies for developing high-performance nonlinear photonic devices in on-chip platforms.

birefringent crystals or quasi-phase-matching (QPM) by periodically engineering the ferroelectric domain of nonlinear crystals have been widely studied and received great success since the 1960s.^[8–10] For the QPM strategy, the domain of the nonlinear ferroelectric crystals is periodically poled by using the electrical field poling method. The resulted $\chi^{(2)}$ nonlinear grating can compensate the wavevector mismatch Δk during the nonlinear process, enabling the continuous power coupling from the fundamental wave to the SHG. In photonic structures that are capable of confining optical fields tightly, e.g., nanophotonic waveguides or microresonators, it is also possible to select diverse modes to achieve PM between the interacting three waves. This leads to an alternative mechanism of SHG, i.e., modal phase matching (MPM).^[11–14]

Nevertheless, the significant mismatch of the different modal profiles blocks the interaction of fundamental and second harmonic waves, resulting in restricted conversion efficiencies of the SHG.

Lithium niobate (LiNbO_3) is a multifunctional crystal that combines a number of excellent features.^[15–21] Recently, thin film crystalline LiNbO_3 (with a typical thickness of 300–900 nm) offers a unique platform to implement on-chip integration of various devices with outstanding performance.^[22–28] The further processing of LNOI wafers enables the construction of nanophotonic waveguides with tightly confined modal fields, in which the interaction of optical waves is significantly enhanced accordingly. Highly efficient nonlinear optical conversion processes have been demonstrated within nanophotonic devices with compact footprints, e.g., loaded waveguide,^[29] ridge waveguide,^[30–36] and microring/disk resonators.^[37–41] Particularly, SHG has been realized in LNOI ridge waveguides through the QPM mechanism, reaching conversion efficiency at the order of $10^3\% \text{ W}^{-1} \text{ cm}^{-2}$.^[30–36]

For $\chi^{(2)}$ nonlinear frequency conversion process within a homogeneous, symmetrical microresonated structure, the selection rule for conservation of parity, which is determined by the spatial overlap integral of the interacting electric field components, must be fulfilled plus the PM condition.^[42,43] However, this selection rule could be exceeded by introducing disturbances of the nonlinear susceptibility to break the symmetry of the interacting

1. Introduction

Quadratic nonlinearity $\chi^{(2)}$ is the fundamental of a large amount of intriguing optical phenomena, which have been widely applied in various aspects of optics and photonics, e.g., frequency conversion of new light,^[1] electrooptic modulation and optical switching,^[2] single-photon generation and detection in quantum information,^[3–5] and optical microscopy.^[6] The physical mechanism behind the coherent $\chi^{(2)}$ parametric process (including SHG) relies on the phase-matching condition, i.e., both the energy and momentum of the interacting photons during the SHG process should simultaneously keep conservation.^[7] To meet the requirements, phase-matching (PM) by using the dispersion of

L. Wang, F. Chen
School of Physics, State Key Laboratory of Crystal Materials
Shandong University
Jinan 250100, China
E-mail: leiwang@sdu.edu.cn; drfchen@sdu.edu.cn

X. Zhang
Key Laboratory of Laser & Infrared System, Ministry of Education
Shandong University
Qingdao 266000, China

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/lpor.202100409>

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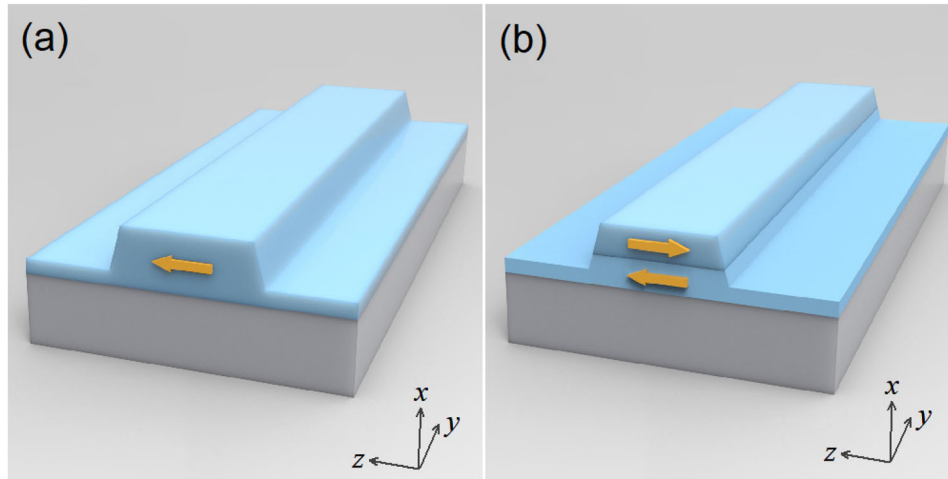


Figure 1. a) Normal ridge waveguide structure in x-cut single-layer LNOI platform and b) the proposed ridge waveguide in double-layer LNOI with reversed polarization.

cross-section. The disturbance can be certain changes of the values or the sign of the $\chi^{(2)}$ susceptibility. For MPM SHG of thin-film LiNbO_3 , Luo et al. implemented an intriguing design of “semi-nonlinear” waveguide by using a TiO_2 thin film (a linear layer) coated on LNOI (a nonlinear layer) ridge waveguide, in which the SHG with high-efficiency of $650\% \text{ W}^{-1} \text{ cm}^{-2}$ has been observed through the MPM.^[44] Here, we propose a new strategy for SHG by using the MPM scheme based on a dual-layer LiNbO_3 thin-film nanophotonic ridge waveguide. Such a structure consists of two layers of x-cut LiNbO_3 thin film, and their polarizations are antiparallel (i.e., two nonlinear layers with 180° difference of polarization vector $\mathbf{P}$). In such a geometry, by selecting suitable modes for fundamental and second harmonic light, a large overlap of the strong confined optical fields as well as MPM could be simultaneously satisfied in a ridge waveguide, enabling an SHG process with ultrahigh conversion efficiency. Under optical pump at the wavelength of 1574.6 nm, we experimentally observe SHG at 787.3 nm with a normalized conversion efficiency of $5540\% \text{ W}^{-1} \text{ cm}^{-2}$. This configuration of dual-layer with two opposite orientations utilizes the two highly nonlinear $\chi^{(2)}$ materials, which is also applicable to many other ferroelectric crystals (e.g., LiTaO_3 , KTiOPO_4), suggesting intriguing potentials for the development of novel nonlinear optical devices with high-performance.

2. Concept Illustration

In ridge nanophotonic waveguides, the light is confined in very small volumes and propagates in regimes of separate guided modes. The profiles of each guided mode (i.e., the electric field portion of the electromagnetic waves) are determined by the geometry of the waveguide cross-sections and the mode order. **Figure 1a,b** shows the schematic plots of a ridge waveguide fabricated on the single-layer and dual-layer x-cut LNOI platform, respectively. Both y-propagating modes of the pump and SH light should be polarized along the z-axis of the LN thin film to utilize the largest d_{33} coefficient ($\approx 27 \text{ pm V}^{-1}$ in the following calculations). For single-layer LNOI, the polarization

is unidirectional and uniform in the waveguide whilst the polarizations are reverse for the bonded dual-layer LN thin films. In the nondepletion regime of the coherent SHG process, the theoretical conversion efficiency of the SHG process is governed by the effective nonlinear susceptibility and the nonlinear modal overlap factor, which is expressed as^[42]

$$\eta = \frac{8\pi^2}{\epsilon_0 c n_1^2 n_2 \lambda^2} \frac{\zeta^2 d_{\text{eff}}^2}{A_{\text{eff}}} \quad (1)$$

where d_{eff} is the absolute value of the effective nonlinear susceptibility in the nonlinear medium, and ζ represents the nonlinear modal overlap factor, given as

$$\zeta = \frac{\int E_{1z}^* E_{1z} E_{2z} dx dz}{\left| \int |E_1|^2 E_1 dx dz \right|^{2/3} \left| \int |E_2|^2 E_2 dx dz \right|^{1/3}} \quad (2)$$

where E_{iz} ($i = 1, 2$) are the electric fields of pump (fundamental frequency light) and SH light modes, respectively. It clearly shows that the mode profile of SH wave should be analogous to that of the pump to ensure a large overlap factor and high conversion efficiency in a homogeneous medium.

Figure 2a shows the calculated refractive indices of quasi-TE modes of the y-propagating ridge waveguides with different widths. The thickness of LiNbO_3 film is $\approx 570 \text{ nm}$. The etched depth is set as 380 nm . The calculated wavelength of the pump/SH light is $1550 \text{ nm}/775 \text{ nm}$, respectively. Two exact PM points are found around ridge widths of 867 nm (I) and 1220 nm (II). **Figure 2b–e** shows the corresponding electric field (E_z) profiles of pump and SH modes at both PM widths, respectively. One can see that the E_z fields of the SH mode in PM point I alter their polarity along the lateral direction, while the E_z fields of PM point II alter the polarity along the depth direction. Electric fields within different mode nodes oscillate in an antiphase manner.

If the intensities of the mode nodes belong to positive/negative polarities nodes are not balanced, a poor overlap factor (ζ)/conversion efficiency is deserved due to the mutually

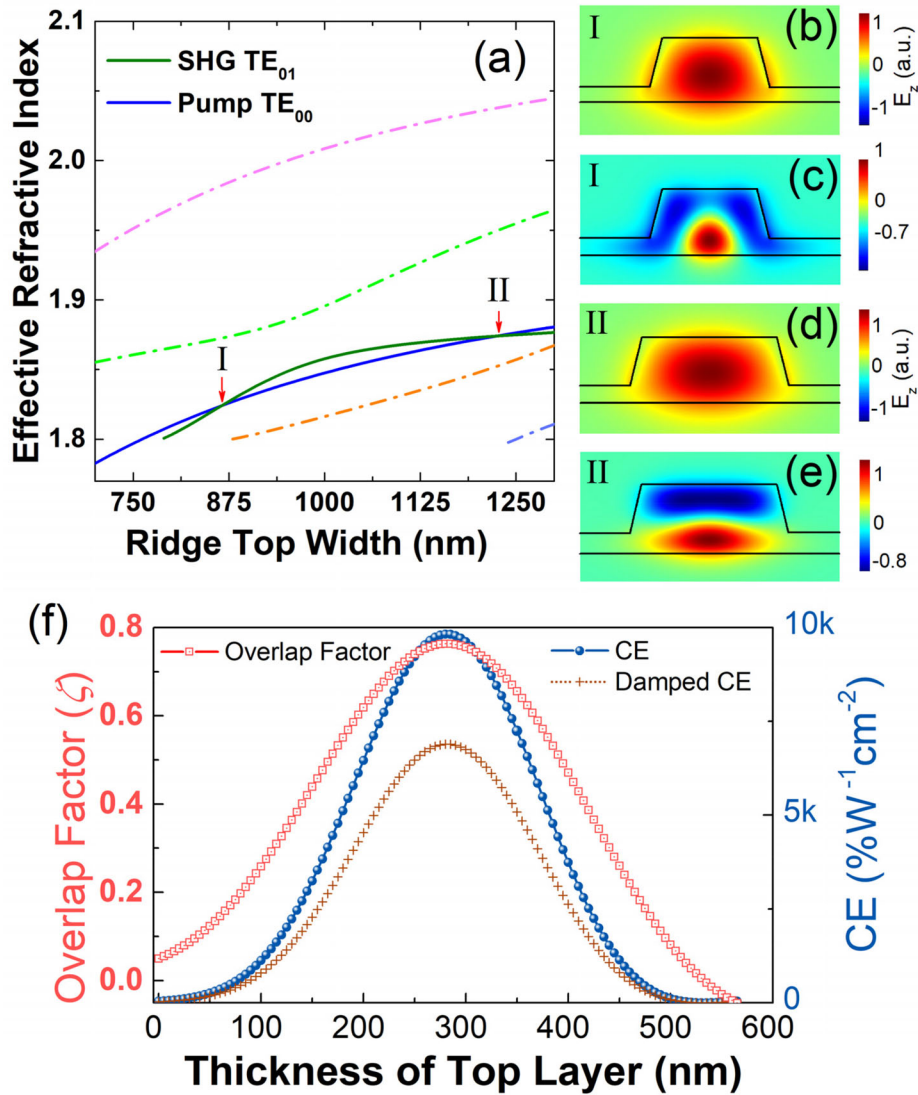


Figure 2. a) Mode effective indices as functions of waveguide top width for TE₀₀ mode of pump and TE modes of SH light. b,c) E_z components of the TE_{00,pump} and TE_{01,SH} at PM I point. d,e) E_z components of the TE_{00,pump} and TE_{01,SH} at PM II point. f) Calculated normalized conversion efficiencies (CE, blue balls; Damped CE, brown crosses) and overlap factor (red squares) of a ridge waveguide as functions of a thickness of top layer for double-layer waveguide at pump wavelength of 1550 nm. CE (blue ball) corresponds to the lossless waveguide. Damped CE (brown cross) corresponds to the lossy waveguide (3.8 dB cm⁻¹ for pump light and 15.2 dB cm⁻¹ for SH light) with a length of 1.2 mm. The top width and thickness of the ridge waveguide are set to be 1230 and 570 nm, respectively.

canceling in Equation (2). In Figure 2e, the light intensities belong to positive/negative polarities (up/downside nodes) are nearly the same, the conversion will be reduced greatly with a ζ value of 0.05. However, if the $\chi^{(2)}$ susceptibility within the up/down nodes region is reversed artificially, the mutually canceling between the side nodes can be reduced greatly. For a ridge waveguide fabricated on the double-layer LiNbO₃ film with reversed polarization (shown in Figure 1b), the nonlinear modal overlap factor can be rewritten as

$$\zeta = \frac{\int d_{\text{Nor}}(x, z) \cdot E_{1z}^* E_{1z} E_{2z} dx dz}{\left| \int |E_1|^2 E_1 dx dz \right|^{2/3} \left| \int |E_2|^2 d_{\text{Nor}}(x, z) \cdot E_2 dx dz \right|^{1/3}} \quad (3)$$

where the $d_{\text{Nor}}(x, z)$ is the normalized space profile of the nonlinear susceptibility. We choose PM point II to demonstrate our proposal. The thickness of two LiNbO₃ layers can be altered to realize the highest ζ factor as well as the conversion efficiency. Figure 2f shows the theoretical conversion efficiency and overlap factor versus the different thicknesses of the top LiNbO₃ film at a pump wavelength of 1550 nm. As one can see, a theoretical maximum normalized conversion efficiency of 9825% W⁻¹ cm⁻² is achieved in a lossless waveguide with a top thickness of 280 nm, corresponding to an overlap factor of 0.76. The damped conversion efficiency is also depicted in Figure 2f as the measured propagation loss of the waveguide is considered. The maximum efficiency is reduced to 6891% W⁻¹ cm⁻² due to the light damping.

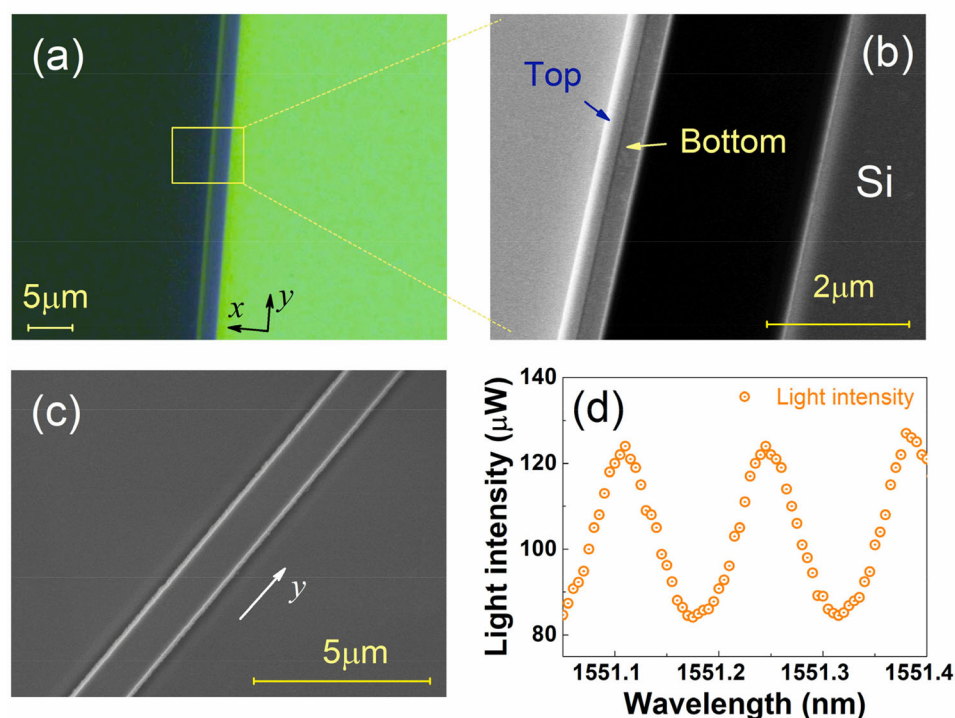


Figure 3. a) The cross section image of the dual-layer LNOI by optical microscope, b) SEM image of the undercut etched end facet of LNOI wafer; Si: the silicon substrate, Top: top LiNbO_3 layer, Bottom: bottom LiNbO_3 layer, and c) SEM image of fabricated ridge waveguide along y -axis. d) The Fabry–Perot interference fringe of ridge waveguide with width of 1230 nm.

It should also be noted that the MPM SHG is eliminated when the thickness of the top layer approaches zero or 570 nm, which corresponds to the case of LNOI with mono LiNbO_3 film.

3. Experimental Results

Figure 3a shows the polished cross-section ($\pm z$ face) of dual-layer LNOI under the optical microscope observation. Due to the optical resolution limit, the double layer cannot be recognized. We etched the polished end face by using HF acid. **Figure 3b** depicts the scanning electron microscopy (SEM) image of the etched end face, showing an evident step profile resulting from the different etching rates belong to the $\pm z$ faces of the thin-film LiNbO_3 . The dark groove results from the undercut etching of the SiO_2 buffer layer. After cutting and cleaning, the ridge nanophotonic waveguides along the y -axis of LiNbO_3 are then fabricated by the standard e-beam lithography (EBL) and Ar^+ ion beam milling.^[12] Finally, the chip is end-polished to be 1.2 mm long for direct light coupling investigation. **Figure 3c** shows the SEM image of the ridge waveguide with a top width of 1230 nm. The Fabry–Perot method is used to evaluate the propagation loss of the ridge nanowaveguide at the telecommunication band.^[45] The obtained interference fringes are shown in **Figure 3d**. The deduced propagation loss is $\approx 3.8 \text{ dB cm}^{-1}$ for the waveguide with a width of 1230 nm.

By tuning the pump light wavelength, we can obtain the SHG efficiency spectrum by using the experimental setup shown in **Figure 4a**. With the propagation loss value, the fiber-to-waveguide coupling loss of 7.4 dB is extracted. **Figure 4d** shows the SHG spectrum as a function of the pump wavelength. The obtained on-

chip normalized conversion efficiency peaks at the wavelength of 1574.6 nm with a full width at half maximum (FWHM) of 3.43 nm, which is 1.08 nm broader than the theoretical SHG spectrum. The difference between the experimental and theoretical curves is likely due to the nonuniformity of the ridge waveguide (variations of width and thickness, see the Supporting Information for more details). By fixing the pump wavelength at 1574.6 nm, the SHG yield as a function of pump power is measured and depicted in the inset of **Figure 4d**, which shows an evident quadratic relation to the pump power. By fitting the experimental data of the inset, we obtain a recorded on-chip conversion efficiency of $5540\% \text{ W}^{-1} \text{ cm}^{-2}$. According to the calculations, the light intensities of the two side nodes shown in **Figure 1e** are nearly the same. Considering the reverse polarization in the two LiNbO_3 layers, the mutually canceling between the two side nodes of $\text{TE}_{03,\text{SHG}}$ will be avoided substantially. The quite high conversion efficiency is also due to the tiny width of the ridge waveguide, which ranges from 1100 to 1300 nm according to the PM wavelength. This width is relatively smaller than the previous work.^[31,32] The compact geometry leads to the small effective area (A_{eff} in Equation (1)) and the high conversion efficiency. It should also be noted that the d_{eff} value of the 1st-QPM process is partially compromised by a factor of $2/\pi$ to the bulk value, whilst the d_{eff} is not reduced during the MPM process between the $\text{TE}_{00,\text{pump}}$ mode and $\text{TE}_{01,\text{SHG}}$ mode. Normally, the conversion between $\text{TE}_{00,\text{pump}}$ and $\text{TE}_{00,\text{SHG}}$ modes during the QPM is preferred because of their similar modal patterns. Our calculation shows an overlap factor ζ value of 0.98 between the $\text{TE}_{00,\text{pump}}$ and $\text{TE}_{00,\text{SHG}}$ modes for the ridge waveguide shown in **Figure 2**.

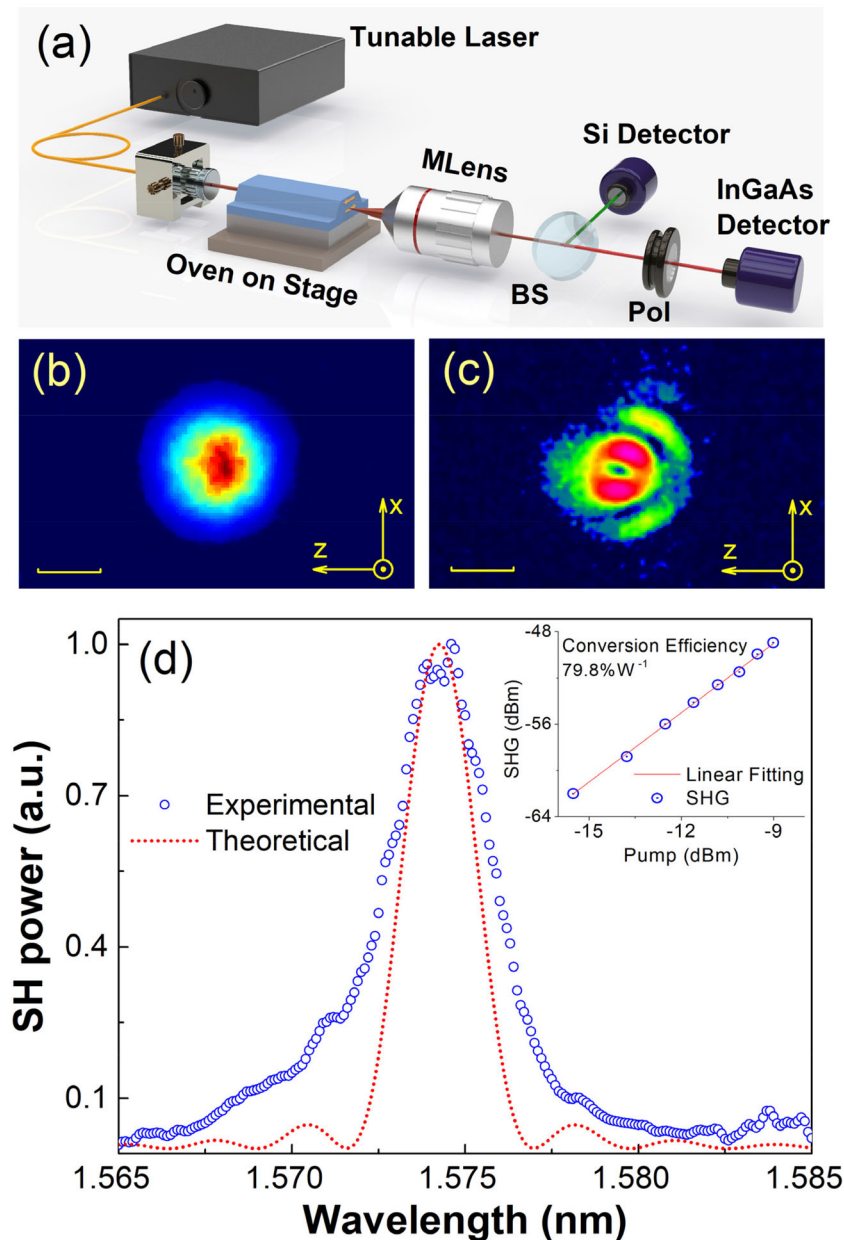


Figure 4. a) Schematic of the optical end-fire coupling setup. Tunable Laser, TSL210VF; MLens, microscope lens; BS, beam splitter; Pol, polarizer; b) the measured pump and c) SHG mode pattern, and d) measured SHG PM curve of dual-layer ridge waveguide with top width of 1230 nm at room temperature. The inset figure shows the SHG power as a function of the pump power. The scale bars in (b) and (c) represent 1 μm .

The ζ factor between $\text{TE}_{00,\text{pump}}$ mode and $\text{TE}_{01,\text{SHG}}$ mode is about 0.76; nevertheless, it is quite acceptable considering the uncompromised d_{eff} value during the exact MPM process.

To show the thermal tuning capability of the nonlinear nanophotonic waveguide, we tune the temperature from 30 to 100 $^{\circ}\text{C}$ and collected the corresponding SHG PM curves. **Figure 5** shows the experimental and simulated PM wavelengths as a function of temperature of the sample. It demonstrates that the SHG spectrum is red-shifted as the device temperature increases. The experimental and simulated tuning slopes are 0.137 nm K^{-1} and of 0.098 nm K^{-1} , respectively. The larger ex-

perimental value of the tuning slope may be due to the thermal expansion and pyroelectric effect, which are sensitive to environment changes in the wavelength-scale waveguides, are not considered in the simulation model.

4. Discussion and Conclusion

It should be pointed out that, despite the high propagation loss, the nano-waveguides based on dual-layer LNOI in this work are still promising because of the ultrahigh normalized conversion efficiency through the MPM mechanism. Another intriguing

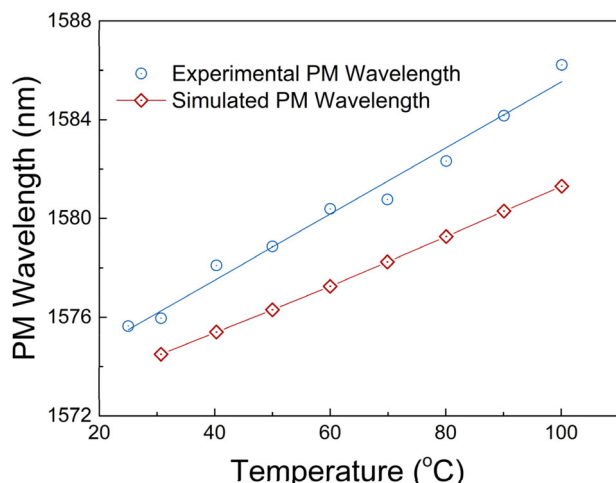


Figure 5. Experimental and simulated PM wavelength as functions of temperature. The simulations take into account the temperature and wavelength dependence of the congruent LiNbO_3 index.

advantage of the dual-layer LNOI is that it avoids the strict requirement for the periodic poling of LiNbO_3 . There are still some possibilities to improve further for our demonstrations. Nevertheless, the urgent solution is to optimize the EBL/dry etching process for the dual-layer LNOI. For example, recent investigations have shown that, by using hydrogen silsesquioxane (HSQ) based dry etching, the propagation loss of LNOI-based photonic devices has been significantly reduced, enabling development of devices with longer interaction length and resonators with high Q factors.^[23] Therefore, in the future work, HSQ masking could be utilized to fabricate ridge waveguides with more smooth sidewalls on the dual-layer LNOI platform. Meanwhile, the single etching process during the HSQ etching could improve

the fidelity of the lithographic pattern transfer and the width control accuracy of waveguides. In addition, the novel chemo-mechanical polishing could also be applied to produce low-loss ridge waveguides on dual-layer LNOI.^[46,47] Grating couplers and tapered waveguide couplers may also be utilized to boost the in-coupling light power and hence the power conversion efficiency. Another option is to use microring resonators on the platform of dual-layer LNOI instead of the straight nanophotonic ridge waveguide due to the fact that the resonance-enhanced nonlinearity heralds the on chip parameter conversion devices towards single-photon level.^[48,49] In most integrated photonic devices, the SH signal in the fundamental mode is preferred due to the convenience of light manipulating/modulating and out-coupling to the lens fiber. In straight waveguides, mode converters with aids of metasurface gratings could be applied to couple the light from the high-order mode to the fundamental mode or vice versa.^[50,51] In resonator devices, the high-order SH signal in the resonators could be coupled to the fundamental mode of a bus waveguide through elaborate design and fabrication of the coupling gap between the resonator and the bus waveguide. The present concept of stacking/bonding two nonlinear films could also be applied to not only the other ferroelectric crystals but also the semiconductor materials that possess the outstanding nonlinear susceptibility in the mid-infrared waveband.

In conclusion, we have proposed a new design for LNOI platform, which consists of two bonded LiNbO_3 films. The reverse polarization of the two films enable the modal phase matching with theoretical normalized conversion efficiency approaching to $10^4\% \text{ W}^{-1} \text{ cm}^{-2}$. An experimental conversion efficiency of $5540\% \text{ W}^{-1} \text{ cm}^{-2}$ is obtained in a straight nanophotonic ridge waveguide, which is comparable with those of recently reported on-chip PPLN waveguides based on QPM configuration.^[31–33] Another further intriguing scheme based on the dual-layer bonded LiNbO_3 films is to realize SHG in microresonators, in which very high efficiency has been achieved in the normal single-layer

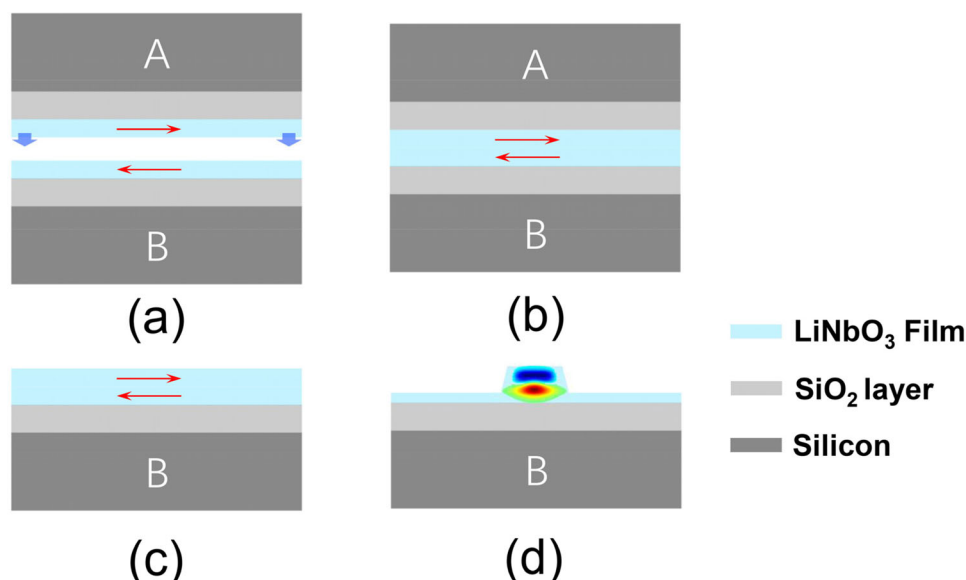


Figure 6. a,b) aligned wafer bonding process of two x-cut LNOI wafers with antiparallel polarizations. c) Removing the top silicon substrate and SiO_2 buffer layer, leaving the dual-layer LiNbO_3 thin films. d) Ridge waveguide fabricated by the EBL/Ar ion milling, showing the mode pattern of SHG light in the ridge waveguide.

LNOI microrings.^[38,39] This approach holds great potentials in many nonlinear optical applications such as efficient on-chip light conversion (e.g., SHG, spontaneous parameter down conversion) and photonic signal detections.

5. Experimental Section

Fabrication of Dual-Layer LNOI and Ridge Nano-Waveguide: The schematic of the fabrication process of the dual-layer LNOI wafer is indicated in **Figure 6**. Two x-cut LNOI wafers with different LiNbO₃ film thicknesses (270 and 300 nm for the top/bottom layers) were cleaned and aligned such that the polarization of the two LiNbO₃ films was opposite to each other. In the next step, two wafers were bonded tightly after the thermal treatment. The top silicon was removed by grinding and wet etching. Then the lower SiO₂ buffer layer was etched away by diluted HF solution, leaving the intact double-layer LiNbO₃ film and the bottom SiO₂/silicon substrate. The ridge nanophotonic waveguides along the y-axis of LiNbO₃ were fabricated by the standard e-beam lithography (EBL) and Ar⁺ ion beam milling. The dry-etched depth was about 380 nm. The remained etching mask was removed, and both facets of the waveguide were then end-polished for the following optical characterizations.

The End-Fire Coupling Setup: The pump light from a continuous-wave tunable laser source (TSL-210VF) was coupled into the waveguide by using a polarization-maintained lensed fiber. The emitted pump/SH light from the rear facet of the waveguide was collected by a 40× microscope lens (NA = 0.65) and separated by a long-pass beam splitter. As the pump light was tuned around the PM wavelength, the generated SH signal can be detected by the silicon detector. The mode patterns were also observed by using the CCD cameras. For the pump light measurements, a polarizer was used to filter the unwanted noise before the InGaAs detector.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

lithium niobate thin films (LNOI), second harmonic generation, dual-layer ridge waveguides, modal phase matching

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